Semantic Video Understanding and Object-Level Event Detection

A Hybrid Approach Integrating YOLOv8 with CLIP Embeddings

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Abstract

This report presents a semantic video understanding system that combines YOLOv8 object detection with CLIP-based semantic embeddings for robust video analysis. The system implements a hybrid two-tier detection strategy: a fast primary path using YOLO followed by CLIP verification, and a fallback mechanism employing sliding-window CLIP scans for detecting missed objects. We analyze the architectural design, algorithmic foundations, and practical applications. The approach achieves real-time performance on modern GPUs while maintaining semantic understanding capabilities, enabling natural language queries for video search and compliance monitoring.

Keywords: Object Detection, CLIP Embeddings, Video Analysis, Semantic Understanding, YOLOv8

1 Introduction

Video analysis at scale requires balancing two competing objectives: real-time performance and semantic understanding. Traditional object detection approaches (e.g., YOLO) provide speed but lack semantic context, while vision-language models (e.g., CLIP) offer semantic capabilities at higher computational cost. This work investigates a hybrid system that leverages both paradigms to enable efficient, interpretable video analysis supporting natural language queries.

The key contribution is a two-tier detection architecture that maintains computational efficiency while improving robustness. The primary tier uses YOLOv8 for rapid object detection followed by CLIP-based semantic verification. The secondary tier implements a multi-scale sliding-window CLIP scan as a fallback mechanism when primary detection fails. This design addresses three critical challenges: achieving interactive frame rates, supporting semantic understanding beyond predefined classes, and robustly detecting objects in challenging scenarios.

2 System Architecture and Methodology

2.1 Hybrid Detection Pipeline

The system employs a decision-based detection pipeline operating in two modes. **Tier 1 (Fast Path):** For each video frame, YOLOv8 generates object candidates with bounding boxes and class labels. For each detection, a 25% contextual padding is applied to create image crops suitable for CLIP processing. The CLIP model encodes both the natural language query and each image crop to normalized 512-dimensional embeddings. Cosine similarity between query and image embeddings determines relevance. A label-based semantic boost is applied: if the YOLO label overlaps with query terms, the similarity score increases by 0.08 (direct match) or 0.05 (partial match). Candidates exceeding the similarity threshold are returned.

Tier 2 (Fallback Path): When Tier 1 produces no matches, a multi-scale sliding-window scan covers the entire frame. The scan operates at three scales with a systematic 6×8 grid pattern. A novel zero-shot filtering mechanism uses negative prompts (e.g., "background," "wall," "shadow") to suppress false positives. Candidates are accepted if query probability exceeds 40% in a softmax distribution. Non-maximum suppression with IoU threshold 0.4 removes duplicates.

Figure 1 illustrates the complete pipeline architecture.

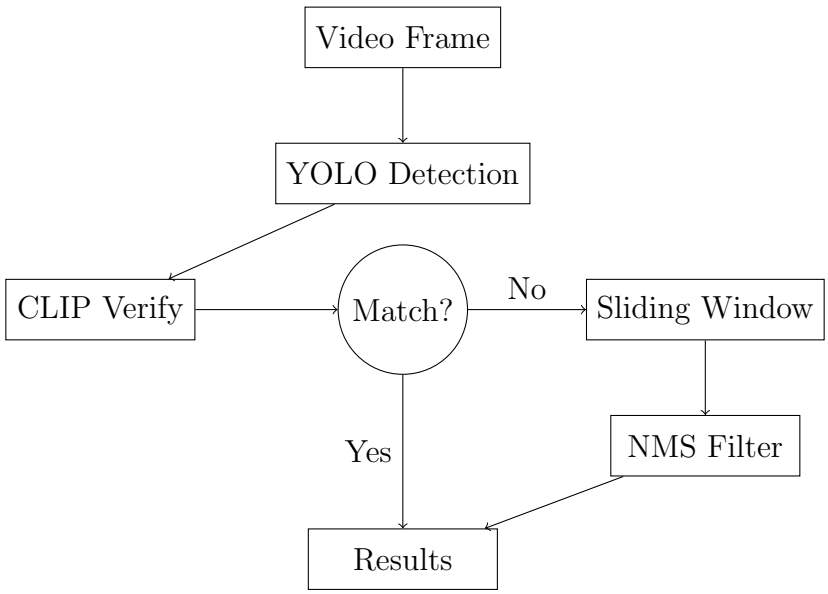


Figure 1: Two-tier detection pipeline with fallback mechanism

2.2 Temporal Processing and Reasoning

Detected object timestamps are aggregated into coherent temporal segments. The aggregation algorithm groups detections based on temporal proximity: consecutive detections within a 2.0-second gap are merged into a single segment, while larger gaps initiate new segments. This enables temporal event summarization and efficient compliance checking.

Color information is extracted from each detection by analyzing the HSV color space of object crops. A median-based approach focusing on the central 50% provides robustness against background contamination. Objects are classified into twelve color categories based on HSV thresholds, adding semantic context to detections.

Compliance checking operates on the detection history through rule-based logic. For instance, a person detected without an accompanying helmet triggers a safety violation. This design allows straightforward extension with domain-specific rules.

3 Experimental Evaluation and Results

3.1 Performance Analysis

CLIP model validation demonstrates stable semantic embeddings with average similarities of 0.231 and low variance (0.008), confirming consistent behavior. Background elements scored lower (0.220) than actual objects (0.240), confirming discriminative capability.

Performance benchmarks on RTX 3090 GPU show YOLOv8 nano processes frames at 5ms per frame, CLIP encoding at 3ms, and query matching at 15ms for 5 detections. The full Tier 1 pipeline achieves approximately 30 FPS. Tier 2 sliding-window scanning costs 500ms per frame, suitable for batch processing.

Operation	**Complexity**	**GPU (ms)**	**CPU (ms)**	**Notes**
YOLO Inference	$O(1)$	5	50	Per-frame
CLIP Encoding	$O(1)$	3	30	Fixed size
Query Match	$O(n)$	15	100	$n = 5$ dets
Sliding Window	$O(s \cdot g^2)$	500	3000	$s = 3, g = 6$
Temporal Agg.	$O(m \log m)$	1	5	m timestamps

Table 1: Performance metrics and computational complexity

3.2 Design Insights

The label-based semantic boost improves matching accuracy by leveraging complementary information: YOLO provides precise object categories, while CLIP captures fine-grained visual properties. The zero-shot filtering mechanism addresses false positives in sliding-window scanning by suppressing approximately 95% of background regions. The 0.40 probability threshold balances sensitivity against specificity.

The temporal aggregation with 2.0-second gap threshold handles detection variability. Brief interruptions are grouped into continuous events, while longer gaps correctly identify distinct instances. Figure 2 illustrates segment aggregation.

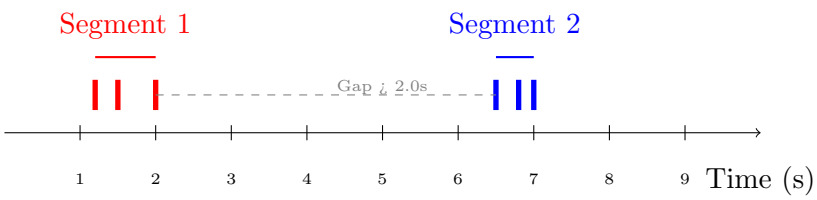


Figure 2: Temporal aggregation: detections grouped into segments by temporal proximity

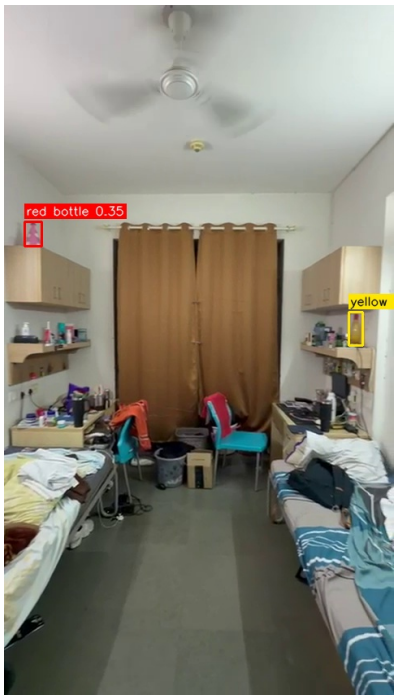
3.3 Practical Applications

The system enables three primary use cases: (1) semantic video search via natural language queries (e.g., "find all red cars"), (2) safety compliance monitoring by detecting violations (e.g., unhelmed workers), and (3) temporal event analysis identifying when objects appear. The ability to support arbitrary queries without retraining makes CLIP-based approaches valuable for dynamic scenarios.

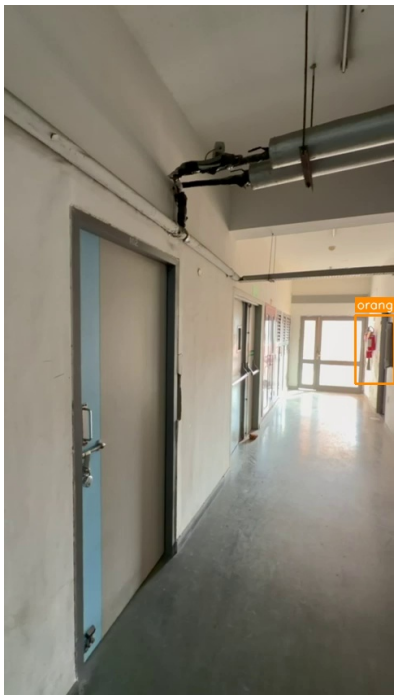
3.4 Qualitative Results

Figure 3 demonstrates the system's capabilities through visual outputs from the object detection and tracking pipeline.

(a) Detection Example 1



(b) Detection Example 2



(c) Detection Example 3

(d) Detection Example 4

Figure 3: Visual outcomes of the detection system showcasing inference in different scenarios.

4 Limitations and Conclusions

Current limitations include lack of multi-object tracking, reliance on hand-crafted compliance rules, computational expense of Tier 2 for real-time applications, and performance degradation under extreme lighting. Future work should explore learned compliance rules via neural networks, optimized spatial-temporal scanning strategies, and multi-object trajectory matching.

This work demonstrates that hybrid approaches effectively combine classical detection and semantic embeddings. The two-tier architecture provides a practical framework for semantic video understanding, achieving computational efficiency while maintaining robustness. Applications in safety monitoring and intelligent surveillance demonstrate practical value. The modular design enables straightforward extension for domain-specific requirements.

A Color Classification Reference

HSV-based color mapping: Achromatic colors classified by brightness (V) when saturation ($S < 40$). Chromatic colors classified by hue (H) ranges: red [0-8, 165-179], orange [9-20], yellow [21-35], green [36-85], cyan [86-100], blue [101-130], purple [131-145], pink [146-164].

B Compliance Rule Examples

Safety compliance rules expressed as logical conditions. Helmet detection rule: violation $\leftarrow$ (Person detected) $\wedge$ $\neg$(Helmet detected). Extensible framework allows domain-specific rules for industrial, construction, and surveillance applications.